1. Mean (Average)

Formula: $\text{Mean} = \frac{\Sigma X}{n}$

$$\bar{X} = \text{Mean} = \frac{18+21+26+27}{4} = \frac{92}{4} = 23$$

2. Standard Deviation

(Age) X	Age-Mean	(Age - Mean)^2
18	18 - 23 = -5	25
21	21 - 23 = -2	4
26	26 - 23 = 3	9
27	27 - 23 = 4	16

$$\sigma = \sqrt{\frac{\Sigma(x-\mu)^2}{N}}$$

$\Sigma(\text{age} - \text{Mean})^2 = 54$

$$\sigma = \sqrt{\frac{54}{4}} = \sqrt{13.5} = 3.67$$

$$X_i' = \frac{X_i - \bar{X}}{\sigma}$$

$$X'_{18} = \frac{18-23}{3.67} = -1.36$$

$$X'_{21} = \frac{21-23}{3.67} = -0.54$$

$$X'_{26} = \frac{26-23}{3.67} = 0.82$$

$$X'_{27} = \frac{27-23}{3.67} = 1.09$$

(A) Min-Max Scaling

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Min = 18 , Max = 27 , Range = 27 - 18 = 9

Age		Min-Max Scaling
X'_{18}	$= \frac{18-18}{9}$	$= 0$
X'_{21}	$= \frac{21-18}{9}$	$= 0.33$
X'_{26}	$= \frac{26-18}{9}$	$= 0.89$
X'_{27}	$= \frac{27-18}{9}$	$= 1$

(B) Mean Normalization

$$X' = \frac{X - X_{Mean}}{X_{max} - X_{min}}$$

Age		Mean Normalization
X'_{18}	$= \frac{18-23}{9}$	$= -0.56$
X'_{21}	$= \frac{21-23}{9}$	$= -0.22$
X'_{26}	$= \frac{26-23}{9}$	$= 0.33$
X'_{27}	$= \frac{27-23}{9}$	$= 0.44$

(c) MaxAbsolute Scaling $X_i' = \frac{X}{|X_{max}|}$

$$X'_{18} = \frac{18}{|27|} = 0.67$$

$$X'_{21} = \frac{21}{|27|} = 0.78$$

$$X'_{26} = \frac{26}{|27|} = 0.96$$

$$X'_{27} = \frac{27}{|27|} = 1$$

(d) Robust Scaling $X_i = \frac{X_i - X_{median}}{IQR}$

$$\text{Median} = \frac{21+26}{2} = \frac{47}{2} = 23.5$$

$$Q_1 = \frac{18+21}{2} = 19.5$$

$$Q_3 = \frac{26+27}{2} = 26.5$$

$$IQR = Q_3 - Q_1 = 26.5 - 19.5 = 7$$

$$X'_{18} = \frac{18-23.5}{7} = -0.79$$

$$X'_{21} = \frac{21-23.5}{7} = -0.36$$

$$X'_{26} = \frac{26-23.5}{7} = 0.36$$

$$X'_{27} = \frac{27-23.5}{7} = 0.50$$